

```

gpg --gen-key - - Run - Windows
Key is valid for? (0) 0
Key does not expire at all
Is this correct? (y/N) y

You need a user ID to identify your key: the software constructs the user
ID
from the Real Name, Comment and E-mail Address in this form:
'Heinrich Heine (Der Dichter) <heinrich@duesseldorf.de>'

Real name: FOSSwire Test
E-mail address: peter@fosswire.com
Comment:
You selected this USER-ID:
'FOSSwire Test <peter@fosswire.com>'

Change (N)ame, (C)omment, (E)-mail or (0)key/(0)uit?

```

Using public key encryption

to encrypt the data, and then must be supplied again to decrypt it. The data without the password is theoretically useless, but anyone with the password can decrypt it.

Public Key encryption uses a different style of algorithm. To begin with, you'll generate two matching "keys"; a public key, and a private key. The characteristic of these keys is such that data encrypted with one can only be decrypted with the other. By generating a public/private key pair, someone can encrypt data using the public key that can only be decrypted using the associated private key. If all you have is the public key, you can't even decrypt what you've just encrypted.

The intended recipient needs to generate a public/private key pair. In the Windows Command Prompt, enter `gpg --gen-key`. First select which kind of key you want, as well as the keysize (you can also accept the default which is 2048 bits). You must also specify when the key will expire and to whose email and name it will be valid. Finally, enter a passphrase to protect your key and GPG will begin compiling a key pair. During the process, it's a good idea to move the mouse or access your drives as this will give the random number generator more info to work with.

At this point your secret key and your public key have been generated, and placed on your key ring (which can be managed via the key ring editor). In order to get the public key to the person who wants to encrypt your data, you'll need to export it: